

SHREYA SAVANT

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Education

Concordia University, Montreal, QC

Expected Graduation: May 2029

Doctor of Philosophy, Information and Systems Engineering

September 2025 – Present

- VoltAge Doctoral Fellowship | Focus: Multi-agent RL, GNNs, distributed optimization for power systems

Master of Applied Science, Quality Systems Engineering

January 2023 – August 2024

- MITACS Accelerate Scholar | Thesis: Memory architectures for LLM dialogue coherence
- Published: Conference on Artificial Intelligence (CAI) 2026, EACL 2026

D. Y. Patil College of Engineering, India

B.Tech, Computer Science

August 2018 – June 2022

Technical Skills

- **Deep Learning & ML:** PyTorch, TensorFlow, scikit-learn; Training large models (distributed training, mixed precision); CNNs, Vision Transformers; Graph Neural Networks; Reinforcement Learning (PPO, DQN), LLMs (GPT-4, Claude, Llama 3, Qwen); Fine-tuning, RLHF, LoRA
- **Time Series & Forecasting:** Statistical modeling; Sensor data & telemetry processing; Anomaly detection; High-frequency time series analysis; Deep learning for sequence modeling
- **Systems Programming:** Rust (async/tokio, serde, WebSocket clients), Python, Go, Bash, Java, C/C++;
- **Data Engineering:** Spark/PySpark, Databricks; Real-time ETL pipelines; Apache Solr, Airflow; PostgreSQL, MongoDB, Redis; Streaming data (Kafka concepts)
- **Infrastructure & MLOps:** Docker, Kubernetes; CI/CD (GitHub Actions); AWS (EC2, S3, Lambda, SageMaker), Azure AI, GCP; Weights & Biases, MLflow, TensorBoard
- **Quantitative/Trading:** Zero-Knowledge Proofs (ZK-SNARKs, circom), Blockchain (Solana, Bitcoin RPC), Market microstructure, Arbitrage detection, Order book analysis, Polymarket API

Experience

Quantitative Systems Engineer (Independent) | Research & Development

November 2025 – Present

- Architected high-frequency arbitrage detection engine in Rust achieving sub-100ms end-to-end latency for cross-chain BTC/Solana opportunities; processes real-time order book data across 5+ exchanges via async WebSocket streams with custom binary serialization
- Implemented Zero-Knowledge proof system using ZK-SNARKs (circom) for privacy-preserving transaction verification; optimized constraint systems reducing proof generation time by 40% through R1CS circuit minimization
- Built Polymarket prediction market data aggregation pipeline analyzing probability dynamics and order flow across 200+ markets; generated automated trading signals from sentiment shifts with 99.7% system uptime
- Designed low-latency streaming data infrastructure handling 10K+ messages/sec using tokio runtime; implemented lock-free concurrent data structures for order book maintenance

Graduate Research Assistant | Concordia University

September 2025 – Present

- Developing multi-agent deep reinforcement learning framework for power grid optimization using PyTorch; modeling 100+ distributed energy resources as cooperative agents with centralized training, decentralized execution (CTDE)
- Building graph neural networks (GraphSAGE, GAT) for power grid topology modeling and cascading failure prediction; training on 50K+ simulated grid states with node-level and graph-level classification
- Implementing distributed optimization algorithms (ADMM, federated learning) for real-time decision-making under uncertainty; deploying models in partnership with Hydro-Québec for production testing
- Processing telemetry data from smart grid sensors (voltage, current, frequency measurements) for anomaly detection and predictive maintenance

Research Team Lead | Innovation Lab, Concordia

January 2024 – January 2026

- Led team of 4 developing production multilingual conversational AI system using RAG architecture with Gemini LLMs, Redis state management, and vector search (Pinecone); achieved 92% coherence across 1K+ multi-turn dialogues
- Engineered automated UI/UX test generation pipeline using GPT-4o with event log analysis; reduced QA cycles by 50% through intelligent test case synthesis from user interaction patterns
- Built AI-assisted grading system using Llama 3 for topic modeling (LDA) and sentiment analysis on long-form student reflections; improved assessment throughput by 60% while maintaining inter-rater reliability of 0.85+
- Fine-tuned domain-specific embeddings for semantic search improving retrieval recall@10 from 0.67 to 0.84

Applied ML Researcher | 7Dish (MITACS Accelerate)

June 2023 – January 2024

- Architected production RAG system using LangChain, GPT-4, and vector databases for nutrition question-answering; deployed on Azure with 99.5% uptime, serving 10K+ queries/month
- Optimized BERT-based semantic search on Databricks improving retrieval accuracy by 21%; fine-tuned embeddings on large text corpora of domain-specific query-document pairs using contrastive learning
- Built real-time ETL pipeline for knowledge base enrichment processing 100K+ nutrition documents; implemented structured output parsing with LLM-driven entity extraction and relation mapping
- Conducted A/B testing on retrieval strategies (BM25, dense retrieval) analyzing precision@k and latency tradeoffs

Software Development Engineer | SpokenWeb, SSHRC/Concordia

June 2024 – January 2026

- Architected automated data ingestion pipelines for cultural heritage metadata using Ruby on Rails and Apache Solr; indexed 5K+ audio recordings across 13 universities with faceted search reducing query latency to <200ms
- Modernized infrastructure migrating to Docker with CI/CD (GitHub Actions); reduced deployment time by 80% and enabled zero-downtime releases
- Lead presentation at Blacklight Summit 2025 and release ([DOI: 10.5281/zenodo.17705608](https://doi.org/10.5281/zenodo.17705608))

Data Science Intern | Visive Inc.

August 2022 – December 2022

- Developed real-time image search engine on AWS using TensorFlow and CNN embeddings; reduced false positive rate by 16% through metric learning and approximate nearest neighbor search (FAISS)
- Built fraud detection pipeline using ensemble methods (random forests, gradient boosting) achieving 92% accuracy and 0.89 AUC-ROC on 1M+ production transactions; reduced false positive rate from 8% to 3%
- Engineered batch ETL pipelines processing daily transactions with optimized PostgreSQL schemas and partitioning strategies

Selected Projects

Comparative Study: CNN vs Transformer Architectures for Medical Image Classification

- Systematic evaluation of 7 SOTA architectures (ResNet, EfficientNet, DenseNet, ViT, Swin, DeiT) on kidney pathology dataset (212K images, 8 classes); analyzed accuracy-efficiency tradeoffs
- Achieved 94.2% test accuracy with DenseNet-169; documented that CNNs outperform transformers on smaller medical imaging datasets with controlled augmentation protocols

Real-Time Attentiveness Detection via Edge Computer Vision

- Led 5-person team developing real-time engagement monitoring using lightweight CNNs and OpenCV; achieved 30 FPS on edge devices (Raspberry Pi 4) through TensorFlow Lite quantization
- Implemented privacy-preserving on-device inference extracting only aggregated engagement metrics without storing raw video